Privacy Router

A Sovereignty-First Privacy Layer for AI Agents

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HOW IT WORKS

Extractor → Judge → Router

Detects sensitive information in prompts, decides whether to mask or process locally, then routes to the appropriate model.

Local (Ministral 3B)

Cloud (Gemini Flash Lite)

Mask & Send

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Problem

AI agents send sensitive data to external LLMs

PROBLEM

AI agents send sensitive data to external LLMs

Existing solutions (OpenAI Privacy Filter) focus on **English/US PII (SSN, ZIP)**. They cannot understand Korean RRN, +82 phone numbers, or research context. The only options are **share all** or **block all**.

EXAMPLE

RRN / PII Leakage

PII such as Korean RRN (13-digit national ID) slips past existing LLM guardrails. Regex **cannot understand context.**

EXAMPLE

Research Secret Leakage

Unpublished research ideas, lab budgets, submission plans — **competitive intelligence** is undetectable by regex.

CONCLUSION

Privacy Router: context-aware prompt firewall

Detect, mask, route — sensitive data stays local, only safe queries reach the cloud. **Data sovereignty for all users.**

Target Users

AI prompts frequently contain sensitive data, yet existing security solutions focus mainly on English PII.

Korean University Researchers

They input unpublished paper ideas, experimental designs, and IRB-related data into LLMs. **Protecting research competitiveness** is critical.

Professionals & Corporate Researchers

They generate prompts containing trade secrets, internal decisions, and budget data. **Corporate secrecy protection** is essential.

Individual Users

Resident Registration Numbers, phone numbers, emails — **PII** gets casually included in everyday prompts.

KEY INSIGHT

Existing solutions (OpenAI Privacy Filter) focus on **English / US PII (SSN, ZIP)**. No system handles Korean RRN, +82 phone numbers, or research secrets.

Architecture

How a user prompt flows through the Privacy Router pipeline

USER
Raw Prompt
"Look up 901212-1234567 please"

→

EXTRACTOR
SLM Detection
Socratic Sensitivity Detection

→

JUDGE
Policy Decision
allow / mask_and_send / ask_to_user

→

ROUTER
Pure Execution Layer (No LLM)
Local / Cloud / User prompt

MASKINGCONTRACT

Deterministic masking with UID-based placeholders. Mask → LLM → Restore (roundtrip-safe).

TWO-PHASE EXTRACTOR

Phase 1: SLM detection. Phase 2: **Critic pass** for review and gap-filling. Deduplication + hallucination filtering.

OBSERVABILITY (PLANNED FOR FUTURE)

OpenTelemetry + Prometheus. Grafana dashboard. **Planned for future.**

Extractor

Socratic method-based detection — Socratic sensitivity detection with SCREAMING_CASE categories, language-agnostic rules with examples like Korean RRN

PII

Personally Identifiable Info

RRN, phone numbers, emails, passport numbers, name + affiliation pairs

```
"901212-1234567" → RESIDENT_REGISTRATION_NUMBER
"010-1234-5678" → CONTACT_NUMBER
```

Research

Research Secrets

Unpublished research, internal decisions, strategies, manufacturing plans

```
"new attention mechanism idea"
→ UNPUBLISHED_RESEARCH_DATA
```

Business

Business Secrets

Credentials, internal URLs, budgets, salary data

```
"API key sk-proj-..." → API_CREDENTIAL
"budget 500M KRW" → BUDGET_INFORMATION
```

Tags use **free-form SCREAMING_CASE** — generated contextually by the SLM using Socratic reasoning, not from a hardcoded list. Korean-specific detection rules for RRN, phone numbers, research secrets, and business secrets.

Judge

Single-Axis Masking Test — "If I replace every sensitive span with [REDACTED], does the user's request still make sense?"

SINGLE-AXIS MASKING TEST

"If I replace every sensitive span with [REDACTED], does the user's request still make sense?"

Verb heuristic determines action: **Creation** → mask_and_send, **Consultation** → mask_and_send, **Interrogation** → process_locally

✓	✕
Yes → **Mask & send to cloud**	No → **Process locally**

```
// Core concept: Single-Axis Masking Test
SENSITIVE_SPANS replaced with [REDACTED] → request still makes sense? → mask_and_send
SENSITIVE_SPANS replaced with [REDACTED] → request broken? → process_locally
```

Router

Pure Execution Layer (No LLM) — Decision table, UID-based masking, hydration

ROUTE 1

External API

Replace sensitive spans with placeholders → send to cloud LLM → restore originals in response

```
mask_text() → Cloud LLM → hydrate_text()
```

ROUTE 2

Local API

When the prompt itself is about sensitive data. **Entire processing stays local.**

```
process_locally() → Local LLM
```

ROUTE 3

Ask to user

When the system cannot decide. **Asks the user for confirmation.**

```
409 → needs_input → user chooses
```

MASKING CONTRACT

Deterministic masking: **UID-based placeholders** (e.g. RESIDENT_REGISTRATION_NUMBER#1) — no brackets. Originals stored encrypted in SQLite. The mask → LLM → restore cycle ensures originals never reach the cloud.

Detection Examples

Real detection results — contextual understanding is the core strength

RESEARCH IDEA DETECTION

Input: "I'm designing a novel loss function that outperforms contrastive learning for multi-modal alignment."

`NOVEL_LOSS_FUNCTION`
`UNPUBLISHED_RESEARCH_DATA`

→ **mask_and_send** (contextual — no trigger keywords)

BUSINESS STRATEGY DETECTION

Input: "We're planning to acquire StartupX for $50M and integrate their proprietary tokenizer into our next release."

`ACQUISITION_PLAN` `FINANCIAL_FIGURE`
`PRODUCT_INTEGRATION`

→ **mask_and_send**

PII DETECTION

Input: "My RRN is 901212-1234567."

`RESIDENT_REGISTRATION_NUMBER`

→ **mask_and_send** (pattern-based)

Key: Contextual detection is the real strength — the SLM understands *meaning*, not just patterns. Tags are free-form SCREAMING_CASE, generated by Socratic reasoning. Confidence: 0.88–0.98

Two-Phase Extraction

Phase 1: Socratic extraction — Phase 2: Critic pass finds what Phase 1 missed

TWO-PHASE EXTRACTOR

Phase 1: Socratic extraction generates SCREAMING_CASE sensitivity categories.
Phase 2: Critic pass reviews results and fills gaps. Deduplication and hallucination filtering merge the final output.

V2 RESULTS

1 → 4
Models hitting 7/7

$0
Added cost

+4
Improved cases

V1 — PATTERN MATCHING

Keyword-Dependent

Relies on explicit cues: "registration number", "decision", "adoption".

Result: Only Gemini Flash Lite detects business/research secrets. All open-source models: **0%**.

V2 — CONTEXTUAL REASONING

Question-Based Inference

The Extractor applies two contextual reasoning questions to every sentence:

"If this sentence appeared in tomorrow's newspaper, would a competitor gain an advantage?"
"If disclosed before publication, would the researcher be harmed?"

These questions enable the SLM to detect sensitive information **without relying on keywords**. The model understands *meaning* — a sentence like "TSMC 3nm process adoption" is classified as a business secret even though no explicit keyword like "secret" or "confidential" appears.

Cost Analysis

$0.075/month — cost-efficient Two-Tier routing

$0.075	46	31	9
Est. monthly cost	Prompts over 7 days	Sensitive detected	Processed locally

TWO-TIER MODEL STACK

Cost-Performance Balance

MODEL	$/1M TOK	V2 SCORE
Ministral 3B	$0.10	7/7
Gemini 3.1 Flash Lite	$0.25	7/7
DeepSeek v4 Flash	$0.10	7/7
Gemma 4 26B-A4B	$0.06	7/7

BEST COST-PERFORMANCE

Gemma 4 26B-A4B

$0.06/1M tok — achieves the same 7/7 full detection as Gemini Flash Lite ($0.25) at **4x lower cost**.

```
# Cost-efficiency ranking
1. Gemma 4 26B-A4B → $0.009/case
2. Ministral 3B → $0.014/case
3. DeepSeek v4 → $0.014/case
4. Gemini Flash Lite → $0.036/case
```

Privacy & Security

Data flow and threat model

DATA FLOW

Sensitive Data Stays Local

- Raw text → **processed only by local SLM**
- Only masked text sent to cloud LLM
- Originals **encrypted in SQLite**
- Restoration performed locally only

THREAT MODEL

Guaranteed Security Properties

- ✓ **Zero sensitive data leakage** — no external transmission without masking
- ✓ **Deterministic masking** — same input → same masking (reproducible)
- ✓ **Auditable** — every decision logged
- ⚠ **SLM hallucination risk** — detection model may miss → mitigated by Critic pass

CORE SECURITY PRINCIPLE

"Only masked data reaches the cloud" — Privacy Router structurally prevents sensitive data from reaching external LLMs. No sensitive information leaves the system without user consent.

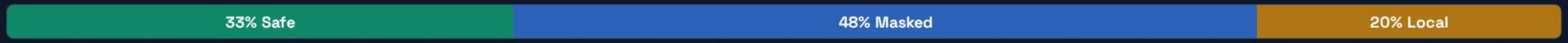
1 / 15

7-Day Usage Log

Real usage data — 46 prompts, 31 sensitive detected, 22 masked, 9 processed locally

46	15	22	9
Total prompts	Safe (sent externally)	Masked & sent	Processed locally

Routing Distribution



DAY 1	DAY 2	DAY 3	DAY 4–7
General conversation	Email drafting (PII)	Paper review	Research tasks
5 prompts	**7 prompts**	**6 prompts**	**28 prompts**

Key finding: Over 7 days, **zero leakage** — not a single sensitive prompt was sent to an external API without masking. 67% of all prompts contained sensitive data.

Tech Stack & Integration

MCP tool + OpenAI-compatible proxy — agent-native privacy

MCP SERVER

process(text, action, model)

Single unified tool. The `action` parameter controls behavior:

```
auto → Full pipeline (detect → route → generate)
classify → Detection only (no LLM call)
generate → Mask + pass to LLM
allow → Skip detection, direct to LLM
```

OPENAI COMPATIBLE

/v1/chat/completions

Drop-in OpenAI SDK proxy. **Any agent** can use Privacy Router by changing the base URL. Built with FastAPI + SQLModel, SvelteKit dashboard, litellm for model routing.

```
# Tech: FastAPI + SQLModel (not SQLAlchemy)
# Frontend: SvelteKit
# Routing: litellm
```

Hermes

Butler agent · MCP direct

OpenCode

CLI agent · Custom provider

A lower bound on model size exists for contextual privacy detection.

Gemma 4 26B crossed that threshold —
at $0.06/1M tokens.

SUMMARY	
Detection Framework	Socratic Sensitivity Detection
Prompt Engineering	v1 → v2: 0% → 100%
Optimal Model	Gemma 4 26B · $0.06/1M
Integration	MCP + OpenAI Proxy
Test Coverage	64 unit + 17 eval
Roadmap	Data → Prompt → Model

DH. Kim & M. Saadati